

LAB : 9

Title: Introduction to Functions in C++

► Objectives

- Understand the concept of functions in C++.
- Learn how to declare, define, and call functions.
- Practice writing functions with and without parameters.
- Apply functions to solve problems using arrays.

► Theory

Functions in C++ allow us to break programs into smaller reusable parts. Each function has:

- A name (identifier).
- Parameters (optional inputs).
- A return type (the kind of value it gives back).

The prototype tells the compiler what the function looks like before it is used.

Syntax:

```
returnType functionName(parameters);
```

Examples:

- `int main();` → returns an integer, no parameters.
- `float divide(int a, int b);` → returns a float, takes two integers.
- `void square(float x);` → returns nothing, takes one float.

► Lab Tasks

Task 1: Function Introduction and Declaration

```
#include <iostream>                // include input/output library

using namespace std;              // use standard namespace


void greet() {                    // function declaration with no parameters
    cout << "Welcome to Lab 09!";  // print greeting message
}


int main() {                      // main function starts
    greet();                      // call the greet function
    return 0;                    // end program
}
```

Task 2: Basic Function Declaration and Call

```
#include <iostream>                // include input/output library

using namespace std;              // use standard namespace


void sayHello() {                 // function declaration with no parameters
    cout << "Hello, World!";      // print hello message
}


int main() {                      // main function starts
```

```
    sayHello();                // call the sayHello function
    return 0;                  // end program
}
```

Task 3: Sum and Average of Array Elements

```
#include <iostream>            // include input/output library

using namespace std;          // use standard namespace


int calculateSum(int arr[], int size) {    // function to calculate sum
    int sum = 0;                // variable to store sum
    for(int i = 0; i < size; i++) {      // loop through array
        sum += arr[i];           // add each element to sum
    }

    return sum;                // return total sum
}

int main() {                    // main function starts
    int numbers[10];            // declare array of size 10
    cout << "Enter 10 integers: ";      // ask user for input
    for(int i = 0; i < 10; i++) {      // loop for 10 inputs
        cin >> numbers[i];           // store each input in array
    }

    int total = calculateSum(numbers, 10); // call function to get sum
    float average = total / 10.0;        // calculate average
    cout << "Sum = " << total << endl;  // display sum
}
```



```
cout << "Average = " << average << endl;    // display average

return 0;                                     // end program
}
```

► Results

- Task 1 displayed: Welcome to Lab 09!
- Task 2 displayed: Hello, World!
- Task 3 correctly calculated the sum and average of 10 integers entered by the user.

► Conclusion

In this lab, I learned how to:

- Declare and define functions in C++.
- Call functions from main().
- Pass parameters to functions and return values.
- Use functions to simplify array operations like sum and average.

Functions make programs modular, reusable, and easier to understand.